

Product Listing Generator : API Lab Report - Nelly Pourmehr

How it works

I pull product images and info from a HuggingFace fashion dataset.

Each image gets turned into base64. I build a prompt with the image plus product details like category, color, usage. That goes to GPT-4o. The model sends back a JSON listing with title, description, features, and keywords. I parse it and save it to a file.

Challenges

Main problem was JSON parsing. GPT-4o wraps its JSON in markdown code fences, even when you tell it not to. That breaks `json.loads()`. I Fixed it with strip the fences before parsing. Honestly , Its weird to see models don't always follow instructions exactly!

Dataset didn't match the lab template either. No `image_path` column, no price column. Images came as PIL objects, not file paths. I needed to encode function to work with PIL directly, and used a placeholder price since the data didn't have one, but afterwards I saw in chat that we can save them locally as well but didn't do it again .

Quality

Results were solid. Titles were catchy, stayed under the character limit. Descriptions hit the word count and picked up real details from the image, like color and pattern. Features and keywords felt specific, not generic.

Happily :D 10 out of 10 products succeeded, no errors.

What I'd improve

Real prices instead of placeholders.

Right now I wait 1 second between each API call. That works for 10 products, but won't hold up at scale. A better approach would slow down only when a call actually fails, instead of always waiting the same amount.

I'd also add a check that goes beyond "is this valid JSON." Right now I only check that the response parses as JSON. I don't check that the fields are actually correct, like making sure "features" is really a list, or "title" isn't empty. That would catch bad output before it reaches the final listing.